

Q#1: Consider that an individual threat agent, like a hacker, can be a factor in more than one threat category. If a hacker breaks into a network, copies a few files, defaces a Web page, and steals credit card numbers, how many different threat categories does the attack fall into?

When a hacker:

- Breaks into a network → Unauthorized Access / Intrusion
- Copies a few files → Information Disclosure / Data Breach
- Defaces a web page → Sabotage / Website Defacement
- Steals credit card numbers → Theft / Financial Fraud

This attack falls into **4 different threat categories**:

1. Unauthorized Access
2. Information Disclosure
3. Sabotage
4. Theft

Q#2: Using the categories of threats mentioned in chapter 02 and the various attacks described, review several cAurrent media sources and identify examples of each threat

Cybersecurity in 2025 faces advanced threats due to evolving tech and geopolitical tensions. Key concerns include:

1. **Sophisticated Ransomware** – Double extortion tactics now target critical sectors like healthcare and finance.
2. **AI-Driven Attacks** – Cybercriminals use AI for phishing, deepfakes, and faster vulnerability exploits.
3. **State-Sponsored Espionage** – Groups like CyberAv3ngers, Brass Typhoon, and Salt Typhoon target infrastructure and telecom for disruption and surveillance.
4. **Quantum Threats** – Future quantum computing may break current encryption; quantum-resistant cryptography is in development.
5. **Supply Chain Attacks** – Hackers exploit third-party links to infiltrate multiple systems.

Recommendations:

- Use AI for threat detection.
- Implement Zero Trust security.
- Secure supply chain partners.
- Prepare for quantum risks.
- Train employees against phishing.

Proactive defense is essential to stay secure in 2025’s digital landscape.

Would you like a one-slide infographic version of this?

Difference: Unconditional vs Computational Security

Aspect	Unconditional Security	Computational Security
Definition	Security holds regardless of the attacker's resources.	Security holds assuming the attacker has limited resources.
Example	One-time pad encryption.	RSA, AES, etc.
Dependence on Time/Power	Not dependent on attacker's time or computational power.	Depends on the assumption that solving the problem takes too long.
Strength	Theoretically unbreakable.	Practically secure (hard to break in reasonable time).
Usage	Rare, impractical for large-scale systems.	Widely used in real-world cryptography.